

**In one line:** VStack builds *every* child now; LazyVStack builds a row only as it nears the viewport (**no reuse**); List reuses platform cells but collects **all ids up front**. **ForEach** **ids** decide which state, animation and callbacks belong to which datum — and since iOS 17/18 the scroll view *reports* position, snapping, geometry and visibility declaratively.

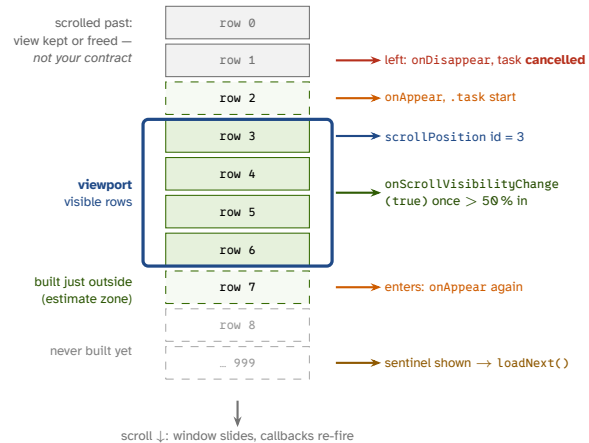
## How it works

- **Creation:** VStack = exact layout, all children loaded at once. LazyV/HStack, LazyV/HGrid (14) create rows on demand; unseen rows' sizes are *estimated*. Apple: plain stack first, lazy when profiling says so.
- **List** visits the whole collection first:  $rows = elements \times views \text{ per element}$  (WWDC23). A variable count (if in ForEach) or AnyView forces it to **build every row** just to count them — filter the *data*, not the *views*.
- **Ids:** Identifiable or id: \.key. id: \.self hashes the whole value  $\Rightarrow$  editing a field = new id = row destroyed + re-inserted. Indices shift on insert/delete  $\Rightarrow$  state lands on the wrong row. Duplicate ids = undefined.
- **Row lifecycle:** onAppear / .task run when a row is shown; onDisappear + *task cancelled* when it leaves; **again on every scroll-back**. Keeping an off-screen row's @State is an implementation detail — durable row state lives in the model, keyed by id.
- **Position** (17): .scrollPosition(id: \$topID, anchor:) — two-way: reads the top-most target's id, write it to scroll; **needs .scrollTargetLayout()** on the stack. 18: ScrollPosition value + .scrollPosition(\$pos): scrollTo(id:), scrollTo(edge: .bottom), viewID(type:). Older: ScrollViewReader.
- **Snapping** (17): .scrollTargetBehavior(.paging) = container-sized pages; .viewAligned = settle on views of the scrollTargetLayout; or your own ScrollTargetBehavior.
- **Observing** (18): onScrollGeometryChange(for:of:action:) maps ScrollGeometry (contentOffset, contentSize, contentInsets, containerSize, visibleRect) to an Equatable; action(old, new) fires **only when that value changes**. onScrollVisibilityChange(threshold: 0.5) per view; onScrollPhaseChange: idle / interacting / decelerating / animating.
- **Look** (17): contentMargins(\_:\_:for: .scrollContent | .scrollIndicators) insets content inside the scroll area (still scrolls edge to edge); scrollTransition { c, phase in }(.topLeading / .identity / .bottomTrailing); visualEffect { c, proxy in } — both return a VisualEffect: **no layout pass**, no GeometryReader.

## Example — snapping, paged feed

```
ScrollView {
  LazyVStack(spacing: 8) {
    ForEach(feed.items) { item in // stable entity id
      Row(item: item)
      .onScrollVisibilityChange { seen in // 18, > 50 %
        if seen { log.impression(item.id) } }
    }
    if feed.hasMore { // sentinel row
      ProgressView()
      .task(id: feed.items.count) { await feed.loadNext() }
    }
  }
  .scrollTargetLayout() // rows = targets
}
.scrollPosition(id: $topID) // 17, read + write
.scrollTargetBehavior(.viewAligned)
.contentMargins(.horizontal, 16, for: .scrollContent)
.onScrollGeometryChange(for: Bool.self) { g in
  g.contentOffset.y > 800 ? action: { _, far in showTop = far }
}
```

## The lazy window



created at first render, 1000 items, 7 visible:

|            |                                              |
|------------|----------------------------------------------|
| VStack     | 1000                                         |
| LazyVStack | ≈ visible + margin; <b>no reuse</b>          |
| List       | cells reused; <b>all 1000 ids read first</b> |

## Traps

- ! ForEach(items, id: \.self) on editable structs, UUID() created in body, or indices — lost row state, wrong row animates, duplicate-id warnings.
- ! if item.visible { Row } inside a List's ForEach, or AnyView rows — every row built to learn the count. Filter the array.
- ! .onAppear { load() } on rows = duplicate requests on scroll-back; guard isLoading + cursor, dedupe appended ids.
- ! Sentinel with plain .task may stay mounted after a page lands — no re-fire; .task(id: items.count).
- ! scrollPosition(id:) without scrollTargetLayout() — binding never updates.
- ! A GeometryReader per row to read offsets — a layout pass each; use onScrollGeometryChange / visualEffect.
- ! onScrollGeometryChange watches the *first* scroll view only.
- ! Transform returns raw contentOffset  $\Rightarrow$  action every frame; map to a Bool / page.

## Remember

**“Lazy builds late, List counts early; ask the scroll view, don't measure it.”**